



Composing Context Compression for Long-Horizon LLM Agents

Controlled Composition and Critical-Fact Retention on top of ACON

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Problem Statement

Problem

How should **history** and **observation** compression be **combined** in a frozen LLM agent, so that we keep the token savings of both **without losing task success**?

Why it matters

- Context grows every step $\Rightarrow$ memory cost, distraction
- ACON's combined setting: **-11.4** points, worse than either stage

Requirements

- Agent weights stay **frozen**
- Same **compression schedule** across settings
- Measure compression quality **directly**

Formulation

Agent step: $a_t = M(h'_{t-1}, o'_t; \theta)$

Pipeline: $\pi = (f_o, f_h, x, \mathcal{K})$ — compressors, input order, trigger steps

$$\max_{\pi} \mathbb{E}[R(\pi)] - \lambda \mathbb{E}[C(\pi)] - \mu \mathbb{E}[L(\pi)]$$

s.t. $\mathbb{E}[R(\pi)] \geq \mathbb{E}[R(\pi_{\emptyset})] - \varepsilon, \quad C(\pi_{ho}) \leq \min\{C(\pi_h), C(\pi_o)\}$

R : task reward C : token cost L : latency

Literature Review (1/2): Existing Methods

ACON [1]

- LLM summarises history and observations past a token threshold
- Guideline optimised in natural language
- **Peak tokens** ↓ **26–54%**, accuracy kept

SAC [2]

- Anchor tokens absorb context via bidirectional attention
- Only anchor KV is kept
- **54.95 F1** at 15× (EPL 51.52)

LLMLingua [4]

- Drops low-information tokens using a small LM
- Task-agnostic, single-step
- Only **39.3%** on AppWorld history

SWE-Pruner [3]

- 0.6B line classifier guided by an agent-written goal hint
- **Tokens** ↓ **23–38%**
- Resolve rate +1.2–1.4 points

MEM1 / MemAgent [5, 6]

- Agent trained with RL to keep a bounded memory
- 8K window → 3.5M-token QA, <10% loss
- **Needs agent weights**

Takeaway

Each method compresses **one** kind of context and is judged only by task success.

Open: how stages *compose*, and whether critical facts survive.

Literature Review (2/2): Comparison

Property	ACON	SAC	SWE-Pruner	LLMLingua	MEM1/MemAgent	Proposed
Output form	Text	KV cache	Text lines	Text tokens	Text memory	Text
Agent weights frozen	Yes	Yes	Yes	Yes	No	Yes
History compression	Yes	No	No	Yes	Merged	Yes
Observation compression	Yes	No	Yes	Yes	Merged	Yes
Composition studied	Partly	No	No	No	No	Yes
Intrinsic quality metric	No	No	No	No	No	Yes (CFR)

- “Partly”: ACON reports a combined setting in an appendix ablation only.
- “Merged”: a single learned memory holds both history and observations.
- Every method compresses *one* kind of context — how the stages **interact** is unstudied.

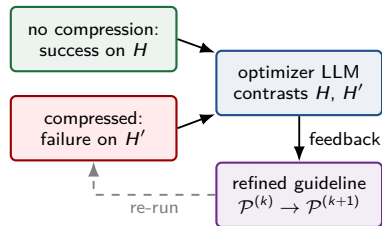
Theory (1/2): ACON — Agent Context Optimization

Threshold-triggered compression

$$h'_t = \begin{cases} f(h_t; \phi, \mathcal{P}_{\text{hist}}) & |h_t| > T_{\text{hist}} \\ h_t & \text{else} \end{cases}$$
$$o'_t = \begin{cases} f(o_t, h_{t-1}; \phi, \mathcal{P}_{\text{obs}}) & |o_t| > T_{\text{obs}} \\ o_t & \text{else} \end{cases}$$

Objective (agent θ frozen)

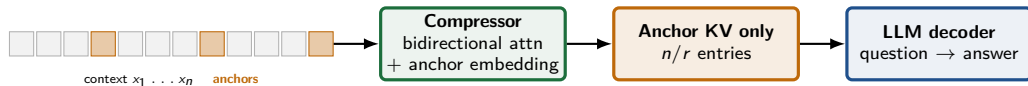
$$\max_{\psi=(\phi, \mathcal{P})} \mathbb{E}[R(s_T(\psi))] - \lambda \mathbb{E}[C(H'(\psi))]$$



Guideline optimisation and distillation

- **UT**: learn from tasks that *fail* after compression $\Rightarrow$ accuracy $\uparrow$
- **CO**: learn from tasks that *succeed* $\Rightarrow$ tokens $\downarrow$; under \$2 per benchmark
- **Distil** gpt-4.1 + $\mathcal{P}^*$ into Qwen3-14B: 99.1% cost cut, >95% accuracy kept

Theory (2/2): SAC — Contextual Semantic Anchors



Mechanism

- Some context tokens are chosen as **anchors** and marked with a learned anchor embedding
- **Bidirectional attention** lets each anchor absorb information from the whole context
- Only the anchors' key-value pairs are passed on: compression ratio r keeps n/r of them

Training and results

- **Autoencoding-free**: no reconstruction objective; trainable adapter pretrained on SlimPajama, fine-tuned on MRQA
- Llama-3.2-1B, 15 $\times$: **54.95** ID / 39.26 OOD F1 (EPL 51.52)
- Output is a **KV cache**, not text — needs model internals

Implementation and Results (1/3): Setup

Hardware and serving

- One NVIDIA RTX 2000 Ada (16 GB)
- Models served locally through Ollama
- Agent qwen2.5:14b, compressor qwen2.5:7b

Benchmarks

- **ACON**: AppWorld train_history_tiny (38 tasks), seed 42, temperature 0
- **SAC**: authors' Llama-3.2-1B checkpoints on MRQA

Deviations from the papers

- Thresholds $T_{\text{hist}}=512$, $T_{\text{obs}}=256$ (paper: 4096/1024, which rarely fire on this split)
- Ollama context of **4096 tokens**, so long baseline prompts were truncated
- One seed per setting so far

What we implemented

- ACON history / observation compression on a fully local stack
- SAC evaluation of the released 5× and 15× checkpoints
- **SAC × ACON pilot**: *FGD* — KL from the full-context model only where it beats the compressed one; *TGA* — question appended to each encoder chunk. 5,000 steps from the 15× adapter, same seed and data order

Implementation and Results (2/3): ACON Reproduction on AppWorld

Run	Tasks	Max iter.	Solved	Avg iter.	Agent input tok.	Wall clock (s)
Baseline, no compression	8	12	1	11.00	269,935	953.9
History compression ($T_{\text{hist}}=512$)	8	12	2	10.75	232,749	882.8
Observation compression ($T_{\text{obs}}=256$)	8	12	1	11.00	261,422	873.6
Baseline, full split	38	30	12 (31.6%)	23.7	2,972,376	7312
Baseline, matched subset	18	30	7	22.22	1,283,906	—
History compression, matched subset	18	30	9	19.50	969,389	—

- Matched 18 tasks: **9 solved vs. 7**, with **24.5% fewer agent input tokens**.
- Not uniform: solved 5 tasks the baseline failed, failed 3 it solved.
- At this scale ACON's reported latency overhead did **not** appear.

Implementation and Results (3/3): SAC and SAC $\times$ ACON

SAC reproduces to within 0.2 F1

15 $\times$: **54.98** ID / **39.25** OOD (published 54.95 / 39.26)

5 $\times$: **63.70** / **47.71** (published 63.63 / 47.72)

Run	ID F1	OOD F1	Δ ID vs. R0 (95% CI)	Δ OOD vs. R0 (95% CI)
R0: SAC fine-tuning (matched control)	51.37	38.62	control	control
R1: + uniform KL (ablation)	51.06	38.17	-0.32 [-1.02, +0.44]	-0.45 [-2.11, +1.44]
R2: + FGD (gated KL, $\lambda=1$)	51.15	38.47	-0.23 [-0.95, +0.47]	-0.14 [-2.04, +1.77]
R3: + TGA + FGD	60.03	44.71	+8.66 [+7.66, +9.60]	+6.10 [+3.62, +8.96]
<i>Published SAC 15$\times$ (20,000 steps)</i>	<i>54.95</i>	<i>39.26</i>		

- **FGD: no measurable effect** — the full-context teacher had higher gold NLL than the student on **90.5%** of examples.
- **TGA + FGD** beats the published checkpoint at a quarter of the steps, but is query-aware (memory not reusable across questions).

Research Gaps

(a) Token savings $\neq$ time savings

- Compressor is itself an LLM call, and rewriting history **resets the KV cache**
- ACON's median time per task: **73 s** (none) $\rightarrow$ **88 s** (history) $\rightarrow$ **102 s** (observation)

(b) Combined setting is schedule-confounded

- Compressing observations changes *when* history compression fires
- ACON's combined drop ($I = -9.5$ points) mixes composition with the shifted schedule

(c) Compression quality is measured only indirectly

- Judged only by task success or F1 — never whether **critical facts** survive
- A failure cannot be traced to information loss vs. agent error

Proposed pipeline

- 1 **Decoupled order:** f_h reads *raw* observations from a side buffer (x^{dec}), so facts dropped by f_o can still reach the summary
- 2 **Controlled schedule:** replay the history-only triggers, $\mathcal{K}(\pi_{ho}) := \mathcal{K}^h$, so I measures composition alone
- 3 **Critical-Fact Retention (CFR):** fraction of task-critical IDs, tokens, paths and numbers kept verbatim, computed offline

Experiments

- Finish the 38-task history run; run observation, combined, paper thresholds and llama3.1:8b compressor
- Log per-call latency $\ell_M, \ell_{f_h}, \ell_{f_o}$, including prefix re-encoding
- Test whether $\rho(\text{CFR}, R)$ predicts success

Stronger evidence

≥ 3 seeds per setting; context window above 4096; for SAC $\times$ ACON, TGA alone and the $5\times/51\times$ ratios.

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Thank You

Questions?

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